

Guided meditation as an adjunct to enhance postoperative recovery after cardiac surgery: study protocol for a prospective randomized controlled feasibility trial.

PMID: 30635064 | Indexed in PubMed

Cardiac surgical procedures are associated with postoperative neurological complications such as cognitive decline and delirium, which can complicate recovery and impair quality of life. Perioperative depression and anxiety may be associated with increased mortality after cardiac surgeries. Surgical prehabilitation is an emerging concept that includes preoperative interventions to potentially reduce postoperative complications. While most current prehabilitation interventions focus on optimizing physical health, mind-body interventions are an area of growing interest. Preoperative mind-body interventions such as Isha Kriya meditation, may hold significant potential to improve postsurgical outcomes. This is a prospective, randomized controlled feasibility trial. A total of 40 adult patients undergoing cardiac surgery will be randomized to one of three study groups. Participants randomized to either of the two intervention groups will receive meditative intervention: (1) commencing two weeks before surgery; or (2) commencing only from the day after surgery. Meditative intervention will last for four weeks after the surgery in these groups. Participants in the third control group will receive the current standard of care with no meditative intervention. All participants will undergo assessments using neurocognitive, sleep, depression, anxiety, and pain questionnaires at various time points in the perioperative period. Blood samples will be collected at baseline, preoperatively, and postoperatively to assess for inflammatory biomarkers. The primary aim of this trial is to assess the feasibility of implementing a perioperative meditative intervention program. Other objectives include studying the effect of meditation on postoperative pain, sleep, psychological wellbeing, cognitive function, and delirium. These will be used to calculate effect size to design future studies. This study serves as the first step towards understanding the feasibility of implementing a mind-body intervention as a prehabilitative intervention to improve postoperative surgical outcomes after cardiac surgery. Clinicaltrials.gov, NCT03198039 . Registered on 23 June 2017.

Incidence and Associated Factors of Postoperative Delirium in Adults Undergoing Cardiac Surgery With Cardiopulmonary Bypass: A Prospective Cohort Study.

PMID: 39654028 | Indexed in PubMed

Delirium is one of the most common and serious complications after cardiac surgery with cardiopulmonary bypass (CPB). A comprehensive assessment of independent risk factors for postoperative delirium (POD) is essential for early detection and prevention. To investigate the incidence and independent associated factors of POD in adults undergoing cardiac surgery with CPB. Prospective cohort design. A total of 203 patients were enrolled in this study from October 2022 to December 2023 in China. Richmond agitation and sedation scale (RASS) and confusion assessment method-intensive care unit (CAM-ICU) were used for assessing delirium symptom. This study analysed various factors for POD, including demographic, physical, psychological, social, spiritual and environmental aspects. Using logistic regression analysis to identify the independent associated factors. A total of 60.1% (n = 122) of patients had POD. Of these cases, 86 (70.5%) were hypoactive delirium, 4 (3.3%) were hyperactive delirium and 32 (26.2%) were mixed delirium. Advanced age (OR = 1.069, 95% confidence interval [CI]: 1.031-1.107; p = 1.000, 95% CI: 1.000-1.001; p < 0.001) were independent predictors of POD. The incidence of POD in patients undergoing cardiac surgery with CPB was high. This study identified advanced age, preoperative depression, postoperative albumin level and duration of mechanical ventilation as significant and

independent predictors of POD. The study's findings highlight the urgent necessity for improved clinical vigilance and proactive management strategies. No patient or public contribution.

Cognitive Training for Reduction of Delirium in Patients Undergoing Cardiac Surgery: A Randomized Clinical Trial.

PMID: 38652478 | Indexed in PubMed

Postoperative delirium is a common and impactful neuropsychiatric complication in patients undergoing coronary artery bypass grafting surgery. Cognitive training may enhance cognitive reserve, thereby reducing postoperative delirium. To determine whether preoperative cognitive training reduces the incidence of delirium in patients undergoing coronary artery bypass grafting. This prospective, single-blind, randomized clinical trial was conducted at 3 university teaching hospitals in southeastern China with enrollment between April 2022 and May 2023. Eligible participants included those scheduled for elective coronary artery bypass grafting who consented and enrolled at least 10 days before surgery. Participating patients were randomly assigned 1:1, stratified by site, to either routine care or cognitive training, which included substantial practice with online tasks designed to enhance cognitive functions including memory, imagination, reasoning, reaction time, attention, and processing speed. The primary outcome was occurrence of delirium during postoperative days 1 to 7 or until hospital discharge, diagnosed using the Confusion Assessment Method or the Confusion Assessment Method for Intensive Care Units. Secondary outcomes were postoperative cognitive dysfunction, delirium characteristics, and all-cause mortality within 30 days following the operation. A total of 218 patients were randomized and 208 (median [IQR] age, 66 [58-70] years; 64 female [30.8%] and 144 male [69.2%]) were included in final analysis, with 102 randomized to cognitive training and 106 randomized to routine care. Of all participants, 95 (45.7%) had only a primary school education and 54 (26.0%) had finished high school. In the cognitive training group, 28 participants (27.5%) developed delirium compared with 46 participants (43.4%) randomized to routine care. Those receiving cognitive training were 57% less likely to develop delirium compared with those receiving routine care (adjusted odds ratio [aOR] 0.43; 95% CI, 0.23-0.77; $P = .007$). Significant differences were observed in the incidence of severe delirium (aOR, 0.46; 95% CI, 0.25-0.82; $P = .01$), median (IQR) duration of delirium (0 [0-1] days for cognitive training vs 0 [0-2] days for routine care; $P = .008$), and median (IQR) number of delirium-positive days (0 [0-1] days for cognitive training vs 0 [0-2] days for routine care; $P = .007$). No other secondary outcomes differed significantly. In this randomized trial of 208 patients undergoing coronary artery bypass grafting, preoperative cognitive training reduced the incidence of postoperative delirium. However, our primary analysis was based on fewer than 75 events and should therefore be considered exploratory and a basis for future larger trials. Chinese Clinical Trial Registry Identifier: ChiCTR2200058243.

Effects of Glucocorticoids on Postoperative Delirium in Adult Patients Undergoing Cardiac Surgery: A Systematic Review and Meta-analysis.

PMID: 34548175 | Indexed in PubMed

Delirium is a common neurologic complication after cardiac surgery and is associated with a poor prognosis. Several studies have explored the effects of glucocorticoids on postoperative delirium (POD). However, conclusion have been inconsistent. The purpose of this systematic review and meta-analysis is

to evaluate the effects of glucocorticoids on POD in adult patients undergoing cardiac surgery. A systematic literature search was conducted using PubMed/MEDLINE, Embase, Cochrane Library/Central, and Web of Science from inception to January 28, 2021. Randomized controlled trials evaluating the effects of perioperative glucocorticoids administration on the incidence of POD in adults (≥ 18 years of age) undergoing cardiac surgery were included. The primary outcome of incidence of POD was assessed using the risk ratio (RR) with a fixed-effects model. Secondary analyses included the severity or duration of delirium, mortality at 30 days, length of hospital and intensive care unit (ICU) stay, duration of mechanical ventilatory support, the incidence of myocardial injury, new atrial fibrillation, renal and respiratory failure, postoperative infection and stroke, and the level of glucose and inflammatory factors. A total 4 trials ($n = 8448$ patients) were included. Glucocorticoids did not significantly reduce the incidence of POD (RR = 0.99; 95% CI, 0.86-1.14) but increased the risk of myocardial injury (RR = 1.22; 95% CI, 1.08-1.38), decreased the duration of mechanical ventilatory support (mean difference, -0.83; 95% CI, -1.32 to -0.34), and led to a tendency toward short length of ICU stay (mean difference, -0.22; 95% CI, -0.47 to -0.03). No significant differences were observed in other secondary outcomes. The perioperative administration of glucocorticoids did not reduce the incidence of POD in adult patients undergoing cardiac surgery but might be associated with shorter duration of mechanical ventilatory support and a tendency toward a shorter length of ICU stay. Furthermore, we found that glucocorticoids may increase the rate of myocardial injury but have no effects on other clinical outcomes. International Prospective Register of Systematic Reviews identifier: CRD42021233458.

Preoperative Vitamin D Level is Associated with Postoperative Delirium After Cardiac Surgery in Patients Over 65 Years of Age.

PMID: 32524976 | Indexed in PubMed

Delirium after cardiac surgery is a devastating and important complication. Delirium is defined as "disturbance in attention (i.e., reduced ability to direct, focus, sustain, and shift attention) and awareness (reduced orientation to the environment)." In this study, we analyzed the association of preoperative vitamin D levels and postoperative delirium after cardiac surgery in patients over 65 years. We retrospectively reviewed the data of 212 adult patients above 65 years of age who underwent isolated coronary artery bypass graft surgery from January 2016 to January 2018. The mean age for Group I was 69.7 ± 7.4 and Group II was 70.6 ± 4.8 years. There were 112 female patients in Group I and 46 female patients in Group II. The patient population was divided into 2 groups based on preoperative serum vitamin D (25-hydroxyvitamin D [25-OHD]) levels (normal range of 25-75nmol/L). Group I included patients with preoperative serum 25-OHD level < 25 nmol/L. Group II included patients with preoperative serum 25-OHD level ≥ 25 nmol/L. The incidence of delirium in this study was 30.2%. In this study, 138 patients (65.1%) had preoperative serum 25-OHD levels < 25 nmol/L, and 74 patients (34.9%) had preoperative serum 25-OHD levels ≥ 25 nmol/L. Preoperative serum 25-OHD levels were associated with postoperative delirium after coronary artery bypass graft surgery. Our retrospective study illustrated that a lower preoperative serum level of 25-OHD was associated with postoperative delirium. Our results showed that 65.1% of patients had preoperative serum 25-OHD levels < 25 nmol/L, and this was associated with postoperative delirium. Vitamin D deficiency exacerbates delirium after coronary artery bypass surgery with cardiopulmonary bypass. Whether the effects of vitamin D deficiency during this event represent separate or interrelated activities with cardiopulmonary bypass is an important question to address and prospective randomized studies are necessary to confirm these results.

Derivation and validation of a preoperative prediction rule for delirium after cardiac surgery.

PMID: 19118253 | Indexed in PubMed

Delirium is a common outcome after cardiac surgery. Delirium prediction rules identify patients at risk for delirium who may benefit from targeted prevention strategies, early identification, and treatment of underlying causes. The purpose of the present prospective study was to develop a prediction rule for delirium in a cardiac surgery cohort and to validate it in an independent cohort. Prospectively, cardiac surgery patients ≥ 60 years of age were enrolled in a derivation sample ($n=122$) and then a validation sample ($n=109$). Beginning on the second postoperative day, patients underwent a standardized daily delirium assessment, and delirium was diagnosed according to the confusion assessment method. Delirium occurred in 63 (52%) of the derivation cohort patients. Multivariable analysis identified 4 variables independently associated with delirium: prior stroke or transient ischemic attack, Mini Mental State Examination score, abnormal serum albumin, and the Geriatric Depression Scale. Points were assigned to each variable: Mini Mental State Examination 4, prior stroke/transient ischemic attack, and abnormal albumin received 1 point each. In the derivation sample, the cumulative incidence of delirium for point levels of 0, 1, 2, and ≥ 3 was 19%, 47%, 63%, and 86%, respectively (C statistic, 0.74). The corresponding incidence of delirium in the validation sample was 18%, 43%, 60%, and 87%, respectively (C statistic, 0.75). Delirium occurs frequently after cardiac surgery. Using 4 preoperative characteristics, clinicians can determine cardiac surgery patients' risk for delirium. Patients at higher delirium risk could be candidates for close postoperative monitoring and interventions to prevent delirium.

Intra-operative hyperoxia and the risk of delirium in elderly patients after cardiac surgery.

PMID: 32305956 | Indexed in PubMed

Delirium is a common complication after cardiac surgery. The aim of our study was to determine the impact of hyperoxia episodes occurring during cardiopulmonary bypass (CBP) on the rate of delirium episodes in the postoperative period. 93 patients, aged ≥ 65 , who underwent elective cardiac surgery (CBP < 90 minutes) were enrolled. The occurrence of delirium episodes was examined every 12 hours for three days after surgery. Eleven patients (11.8%) developed postoperative delirium (POD (+)) and 83 did not (POD (-)). More incidences of severe hyperoxia ($\text{PaO}_2 \geq 26.6 \text{ kPa}$) during CBP were observed in the POD (+) group: 64% had ≥ 2 episodes of hyperoxia, 27% ≥ 3 , and 18% ≥ 4 , while in the POD (-) group: 42%, 13% and 1%, respectively ($P=0.02$). Patients in the POD (+) group had a higher maximum PaO_2 during CBP than the POD (-) group (37 ± 5.8 vs $31.6 \pm 6.6 \text{ kPa}$; $P=0.01$) and a higher mean PaO_2 (30.1 ± 4.5 vs $26.1 \pm 5.6 \text{ kPa}$; $P=0.01$). The optimal maximum PaO_2 cut-off point for the occurrence of delirium was 33.2 kPa (AUC 0.72, $P=0.001$, sensitivity 75%, specificity 38%). We conclude that CBP hyperoxia episodes may be a risk factor associated with the occurrence of postoperative delirium.

The effect of peri-operative dexmedetomidine on the incidence of postoperative delirium in cardiac and non-cardiac surgical patients: a randomised, double-blind placebo-controlled trial.

PMID: 33960404 | Indexed in PubMed

Delirium occurs commonly following major non-cardiac and cardiac surgery and is associated with: postoperative mortality; postoperative neurocognitive dysfunction; increased length of hospital stay; and major postoperative complications and morbidity. The aim of this study was to investigate the effect of peri-operative administration of dexmedetomidine on the incidence of postoperative delirium in non-cardiac and cardiac surgical patients. In this randomised, double-blind placebo-controlled trial we included 63 patients aged ≥ 60 years undergoing major open abdominal surgery or coronary artery bypass graft surgery with cardiopulmonary bypass. The primary outcome was the incidence of postoperative delirium, as screened for with the Confusion Assessment Method. Delirium assessment was performed twice daily until postoperative day 5, at the time of discharge from hospital or until postoperative day 14. We found that dexmedetomidine was associated with a reduced incidence of postoperative delirium within the first 5 postoperative days, 43.8% vs. 17.9%, $p = 0.038$. Severity of delirium, screened with the Intensive Care Delirium Screening Checklist, was comparable in both groups, with a mean maximum score of 1.54 vs. 1.68, $p = 0.767$. No patients in the dexmedetomidine group died while five (15.6%) patients in the placebo group died, $p = 0.029$. For patients aged ≥ 60 years undergoing major cardiac or non-cardiac surgery, we conclude that the peri-operative administration of dexmedetomidine is associated with a lower incidence of postoperative delirium.

A randomized control trial comparing prophylactic dexmedetomidine versus clonidine on rates and duration of delirium in older adult patients undergoing coronary artery bypass grafting.

PMID: 31668468 | Indexed in PubMed

Postoperative delirium occurs in 20-50% of elderly patients undergoing cardiac surgery and increases morbidity and mortality. We investigated whether prophylactic dexmedetomidine could reduce delirium incidence in elderly patients after coronary artery bypass grafting (CABG), compared with clonidine. Prospective observational trial. Academic university hospital. Patients (60-70 years old) who underwent CABG and received either dexmedetomidine or clonidine infusion postoperatively. Patients were randomly allocated to dexmedetomidine or clonidine groups. In the dexmedetomidine group, patients received an initial infusion of 0.7-1.2 $\mu\text{g/kg/h}$; sedation and analgesia were evaluated after 45-60 min. If the Richmond assessment sedation score (RASS) increased from +1 to +4, the infusion rate was increased by 0.1-0.2 $\mu\text{g/kg/h}$ every 30 min, up to 1-1.4 $\mu\text{g/kg}$ body-weight/h. Dexmedetomidine infusion was not discontinued pre-extubation; thereafter, infusion was reduced by 0.1 $\mu\text{g/kg/h}$ until 0.2 $\mu\text{g/kg/h}$. The maximum infusion duration was 72 h. In the clonidine group, patients received an initial infusion of 0.5 $\mu\text{g/kg}$, followed by 1-2 $\mu\text{g/kg/h}$, if the RASS changed from +1 to +4. This was continued throughout mechanical ventilation. Patients were followed up to 5 days post-surgery. Delirium incidence, extubation time, lengths of intensive care unit (ICU) and hospital stay, need for inotropic support or vasopressors, mean arterial blood pressure and heart rate, hospital mortality rate, total postoperative morphine dose, number of patients receiving haloperidol, and adverse events were recorded. Two-hundred-and-eighty-six patients (dexmedetomidine, 144; clonidine, 142) were studied. Dexmedetomidine was associated with lower risk and duration of delirium, shorter mechanical ventilation duration and ICU stay, lower mortality rate, and lower morphine consumption than the clonidine group. Dexmedetomidine significantly decreased heart rates after ICU admission. Postoperative infusion of dexmedetomidine provides a feasible option for postoperative control of delirium after CABG in adult patients.

Effects of cognitive training on cognitive function in patients after cardiac surgery: A systematic review and meta-analysis of randomized controlled trials.

PMID: 39495993 | Indexed in PubMed

Postoperative cognitive deficits frequently occur in patients undergoing cardiac surgery, leaving them with reduced cognitive function. Cognitive training has been shown to improve cognitive function, however, the role in patients after cardiac surgery is unclear. In this study, we aimed to evaluate the effectiveness and safety of cognitive training in patients undergoing cardiac surgery. A systematic search of PubMed, Embase, Cochrane Library, CINAHL, Ovid Medline, Web of Science, CNKI, and Wanfang was conducted until March 2024. The risk of bias was assessed using the Cochrane Risk of Bias Tool. Data were meta-analyzed using RevMan 5.4 software. Potential bias and reliability of evidence were fairly assessed by using the Cochrane risk of bias method and the GRADE evidence grading method. A total of 16 studies involving 1335 cardiac surgery patients were included in this study. Compared with the control group, the cognitive training group had a significantly lower incidence of postoperative cognitive dysfunction (RR 0.35, 95% CI 0.18-0.65, $P = .001$), significantly improved cognitive function (MD 2.54, 95% CI 1.27-3.81, $P < .001$), and a significantly higher quality of life-mental component (MD 5.22, 95% CI 2.32-8.13, $P < .001$), anxiety (MD -6.05, 95% CI -10.96 to -1.15, $P = .02$) and depression (MD -3.97, 95% CI -7.15 to -0.80, $P = .01$) were significantly improved between groups. However, the differences were not statistically significant for postoperative delirium (RR 1, 95% CI 0.38-2.65, $P = 1.00$) and postoperative hospitalization (MD -0.95, 95% CI -2.90 to 1.00, $P = .34$). The present study, based on a low to moderate quality of evidence, suggests that cognitive training improves cognitive functioning, reduces the incidence of postoperative cognitive dysfunction, and has a positive impact on anxiety and depression in patients undergoing cardiac surgery. However, current evidence does not allow for the determination of effects on quality of life, postoperative delirium, and postoperative length of stay.